

Determinants

✓ We list 3 desirable axioms for a f^n

$\det : F^{n \times n} \rightarrow F$ to be called as determinant. We will argue that there is a unique such f^n .

✓ 3 desirable axioms for a determinant:

$$\det : F^{n \times n} \rightarrow F, \quad F \in \{\mathbb{R}, \mathbb{C}\}.$$

① Normalization:
 $\det I = 1.$

② Antisymmetry:

The determinant changes sign when any two rows are interchanged.

Example: $\det [e_{rs}(A)] = -\det A$, e_{rs} is elementary row op. that interchanges rows 'r' & 's'.

③ Linearity in each row:

Determinant is linear in each row separately.

Example: $A_{n \times n} = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_n \end{bmatrix}$

where $A_i = \sum_j c_{ij} \beta_j$,
for some $i \in \{1, \dots, n\}$,
and β_j is j^{th} row of A
 β_i is $1 \times n$ row matrix,
 $c_i \in F$.

$$\det A = \sum_i c_i \det \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ \beta_i \\ \vdots \\ A_n \end{bmatrix}.$$

Note that, $\det[A+B] \neq \det A + \det B$;
 $\det[\lambda A] \neq \lambda \det A$, $\lambda \in F$.

For $A_{n \times n}$, and $\lambda \in F$,
 $\det[\lambda A] = \lambda^n \det A$.

The above 3 axioms give rise to a unique determinant f^n (or a formula).

We now prove some properties of the determinant that follows from the axioms.

Properties:

(a) If two rows of A are equal, then $\det A = 0$.

Proof: Say rows ' r ' & ' s ' are equal. Then,
 $\det(A) = \det[e_{rs}(A)] = -\det(A) \Rightarrow \det(A) = 0$.

(b) If A has a zero row, then $\det A = 0$.

Proof: let row ' i ' be a zero row.

$\det A = \sum_i 0 \cdot \det \begin{bmatrix} A_1 \\ \vdots \\ \beta_i \\ \vdots \\ A_n \end{bmatrix}$ where $\beta_i = [0 \dots 0]$ is $1 \times n$ zero row matrix.

(c) If ~~the~~ ^{= 0} rows of A are linearly dependent, then $\det A = 0$.

Proof: $A_{n \times n} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}$, β_i for $i \in \{1, \dots, n\}$ are $1 \times n$ row matrices

Without loss of generality, say
 $\beta_1 = \sum_{i=2}^n c_i \beta_i$ for $c_i \in F, i \in \{1, \dots, n\}$.

$$\det A = \sum_{i=2}^n c_i \det \begin{bmatrix} \beta_i \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} = 0.$$

(d) Adding or subtracting a multiple of one row from another row leaves the determinant invariant.

Proof: Say c times r th row of A is added to row s of A .

$$\det A = \det \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_r \\ \vdots \\ \beta_s \\ \vdots \\ \beta_n \end{bmatrix} + c \det \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_r \\ \vdots \\ \beta_r \\ \vdots \\ \beta_n \end{bmatrix}$$

$\xrightarrow{r\text{th row}}$
 $\xrightarrow{s\text{th row}}$

$$= \det \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_r \\ \vdots \\ \beta_s + c\beta_r \\ \vdots \\ \beta_n \end{bmatrix} = \det (e_{s \rightarrow s+cr}(A))$$

$= 0$

where

$$A = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix},$$

β_i is $1 \times n$ row matrix for each $i \in \{1, \dots, n\}$.

More generally, if a row is added w/ a linear combination of other rows then determinant remains invariant.

② $\det A = \text{product of diagonal entries}$,
i.e., $\det A = \prod_i A_{ii}$, if A is a diagonal matrix or a Δ gular matrix (upper or lower).

$$A = \begin{bmatrix} A_{11} & & 0 \\ & \ddots & \\ 0 & & A_{nn} \end{bmatrix}, \text{ then } \det A = \prod_{i=1}^n A_{ii}.$$

③ Computation of a determinant formula:

For 2x2 matrices: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\begin{aligned} \det A &= \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix} \\ &= \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} -a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix} \\ &= 0 + ad + 0 + 0 = ad - bc. \end{aligned}$$

Each row splits into n coordinate directions, so the expansion has n^n terms. Most (except $n!$) of those terms w/ automatically vanish. When two rows are in same direction, $\det$ is zero.

Attention to those when the rows point in different directions.

$$\begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} = 0 = \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}$$

For 3×3 matrices:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\det A = \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

↓
decompose into
9 more terms

↓
9 more terms

↓
9 more terms.

$$= \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & 0 \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ a_{21} & 0 & 0 \\ 0 & 0 & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ 0 & 0 & a_{23} \\ a_{31} & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ a_{21} & 0 & 0 \\ 0 & a_{32} & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & 0 \\ a_{31} & 0 & 0 \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & 0 & 0 \\ a_{31} & 0 & 0 \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & 0 & 0 \\ 0 & a_{32} & 0 \end{vmatrix} + \dots = 0.$$

All but $n!$ det's are 0 because a row direction is repeated. There are n choices for a_{ij} entry in the 1st row, $(n-1)$ remaining choices for a_{ij} entry in 2nd row, ..., 1 choice for the last row.

$$\det A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + a_{11}a_{23}a_{32} \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} + a_{12}a_{21}a_{33} \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} + a_{12}a_{23}a_{31} \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix} + a_{13}a_{21}a_{32} \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} + a_{13}a_{22}a_{31} \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix}$$

$$= a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(a_{23}a_{31} - a_{21}a_{33}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

$$= \sum_{\sigma \in S_3} a_{1\sigma(1)} a_{2\sigma(2)} a_{3\sigma(3)} \text{sign}(\sigma),$$

where $S_3 = \text{set of all permutations on } \{1, 2, 3\}$.
 $= \{(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)\}$

$\text{sign}(\sigma) = \begin{cases} 1 & \text{if } N_\sigma \text{ is even} \\ -1 & \text{if } N_\sigma \text{ is odd} \end{cases}$, $N_\sigma = \text{no. of pairwise exchanges reqd. to transform } \sigma \text{ to } (1, 2, 3)$.

For $n \times n$ matrix:

$$A = [a_{ij}]_{\substack{i \in \{1, \dots, n\} \\ j \in \{1, \dots, n\}}} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}, \quad \beta_i = (a_{i1}, \dots, a_{in}) = \sum_{j=1}^n a_{ij} e_j,$$

$\{e_1, \dots, e_n\}$ - ONB

$$\det A = \det(\beta_1, \dots, \beta_n)$$

$$= \sum_{j=1}^n a_{1j} \det(e_j, \beta_2, \dots, \beta_n)$$

$$= \sum_{j_1=1}^n \sum_{j_2=1}^n \dots \sum_{j_n=1}^n a_{1j_1} a_{2j_2} \dots a_{nj_n} \det(e_{j_1}, e_{j_2}, \dots, e_{j_n})$$

Many terms in this sum are 0. In particular, if any two of indices $j_1, \dots, j_n$ coincide,

$\det(e_{j_1}, \dots, e_{j_n}) = 0$. Excluding the zero terms,

$$\det A = \sum_{\sigma \in S_n} a_{1\sigma(1)} a_{2\sigma(2)} \dots a_{n\sigma(n)} \det(e_{\sigma(1)}, \dots, e_{\sigma(n)}),$$

$S_n = \text{set of all permutations on } \{1, \dots, n\}$,
 $|S_n| = n!$.

Notice that, $\det(e_{1\sigma(1)}, e_{2\sigma(2)}, \dots, e_{n\sigma(n)}) = \sin(\sigma)$.

$$\Rightarrow \det A = \sum_{\sigma \in S_n} a_{1\sigma(1)} a_{2\sigma(2)} \dots a_{n\sigma(n)} \text{Sign}(\sigma).$$

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} c_{11} & c_{21} & c_{31} & \dots \\ c_{12} & c_{22} & c_{32} & \dots \\ \vdots & \vdots & \vdots & \\ c_{1n} & c_{2n} & c_{3n} & \dots \end{bmatrix},$$

c_{ij} is cofactor of $(i,j)^{\text{th}}$ entry defined as
 $c_{ij} = (-1)^{i+j} M_{ij}$, M_{ij} = submatrix of A obtained by removing i^{th} row & j^{th} col.

$$A^{-1} = \frac{C^T}{\det A}. \quad (A^{-1})_{ij} = \frac{c_{ji}}{\det A}.$$